SOC Home Lab / Phase 2

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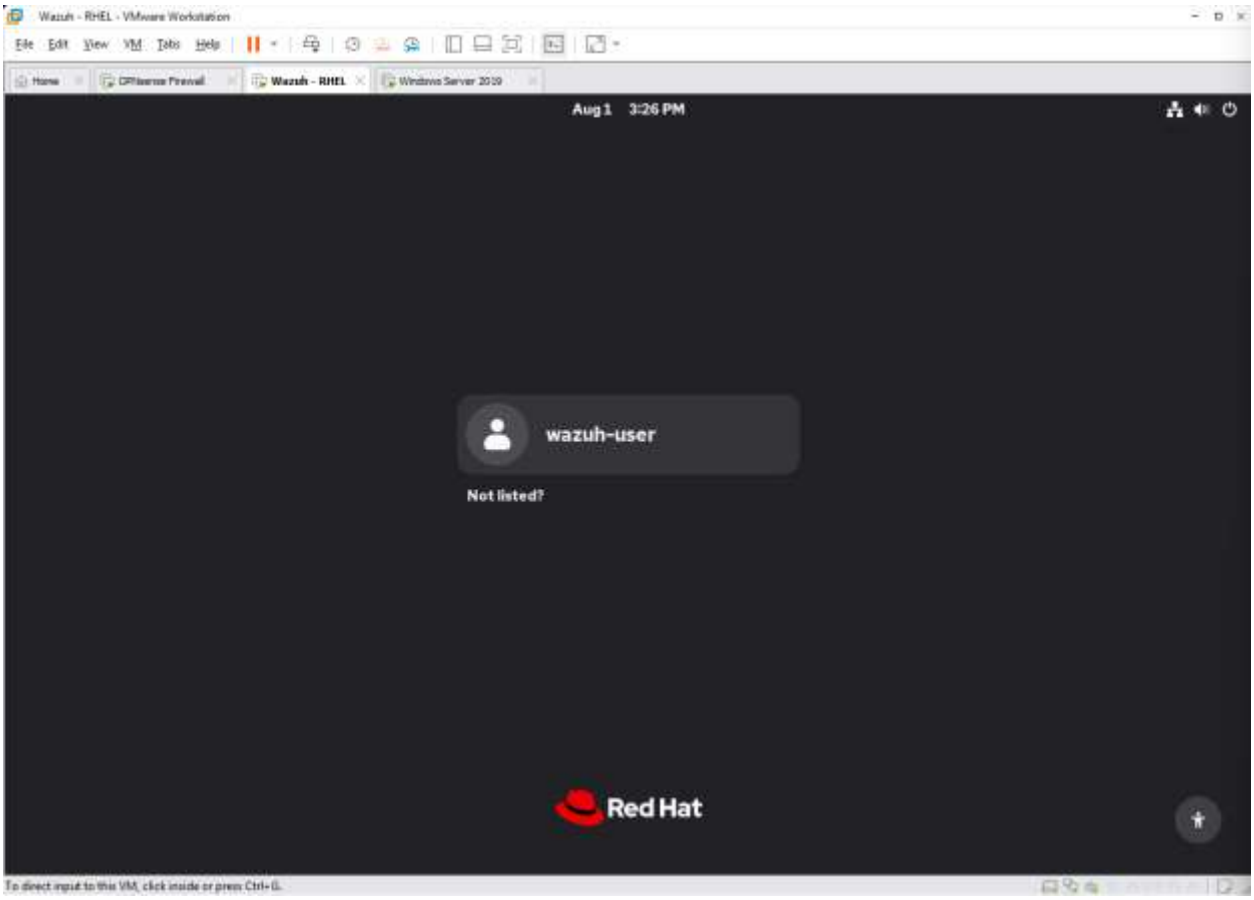
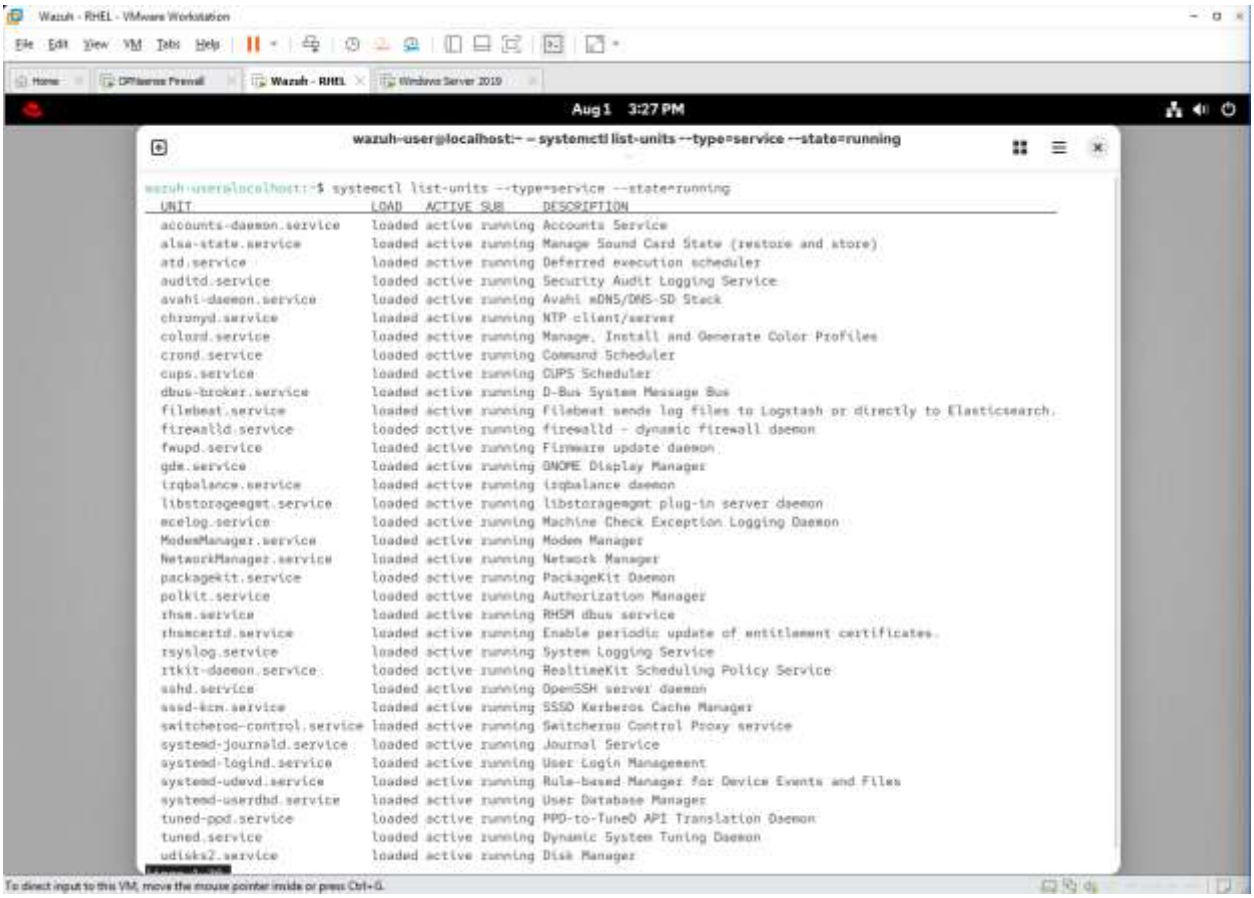
Objective: The objective of phase two is to deploy Wazuh agents to the following systems: Windows Domain Controller (Windows Server 2019 VM) and Windows 10 Workstation. I am using an open-source SIEM (Wazuh) to gain hands-on experience with centralized log aggregation, endpoint telemetry, and rule management.

Deploying Wazuh Agents: Wazuh comes with a few ways to deploy agents but the one I found easiest was using the web UI to generate a command you can copy and paste into your virtual machine's terminal. I first boot up my RHEL 10 VM and verify the Wazuh services are running properly before I head to the web UI.



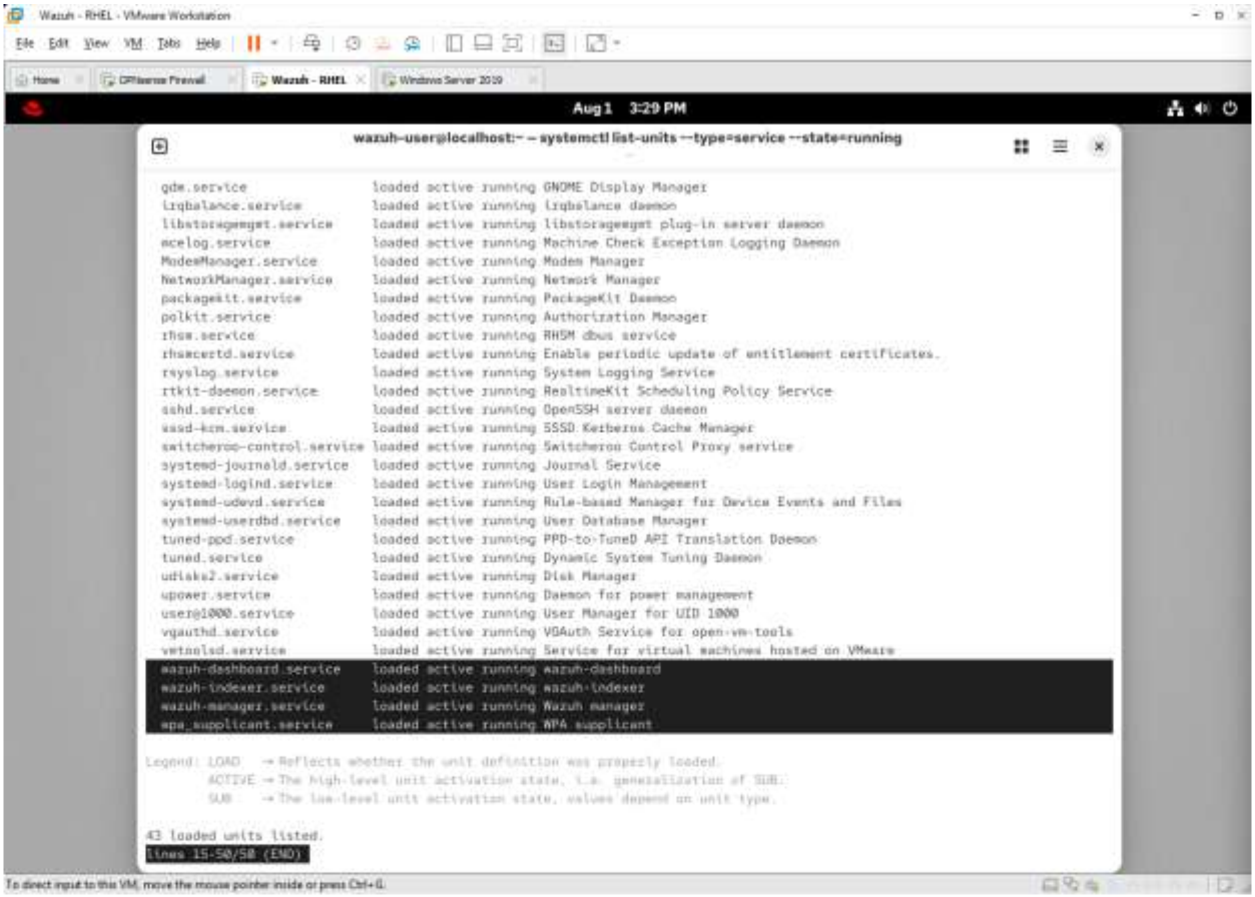
Here I listed all active running services on my RHEL system and proceeded to search for any services with the name "***Wazuh***".



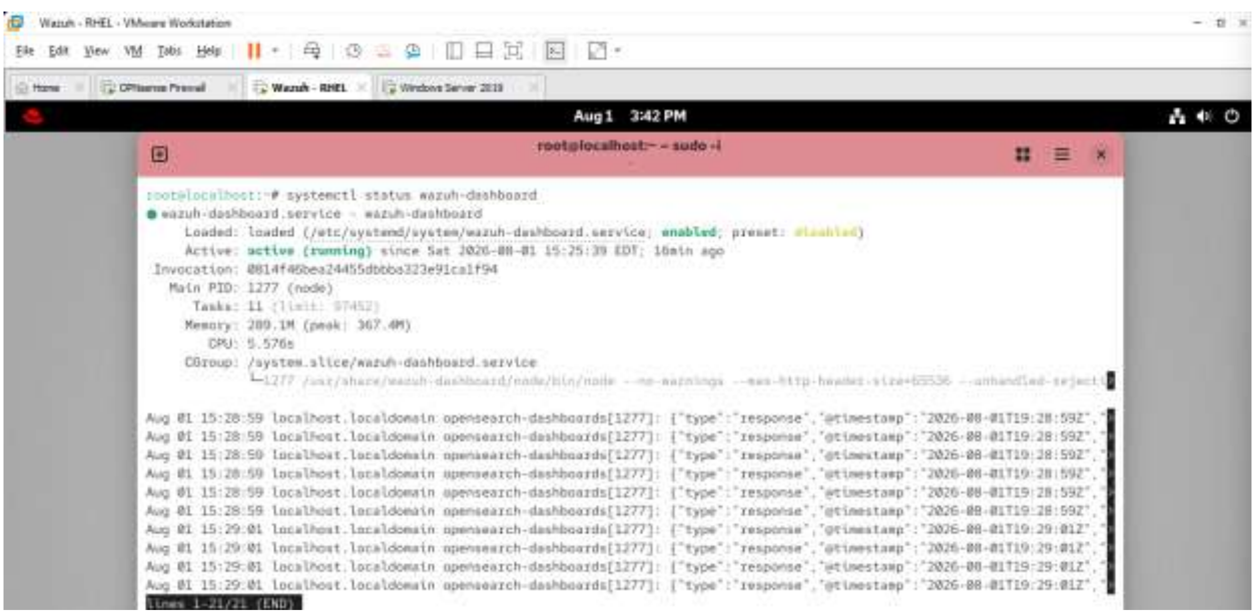
It's important to note that Wazuh relies on three core services working together to have a fully functional SIEM. The three core services are as follows:

- **Wazuh-Dashboard.service** – The web-based user interface. It queries the Indexer to display visual dashboards, threat analytics, system alerts, and platform management tools.
- **Wazuh-Indexer.service** – This service is a high-performance search and storage engine. It indexes processed security alerts from the manager.
- **Wazuh-Manager.service** – The core processing engine ("the brain"). It collects raw telemetry from agents, parses it through decoders and detection rules, triggers alerts, and forwards the processed events to the Indexer.

Here you can see that all the required services are running as expected but if you would like to inspect the services in more detail you can use the command "***systemctl status <service-name>".***



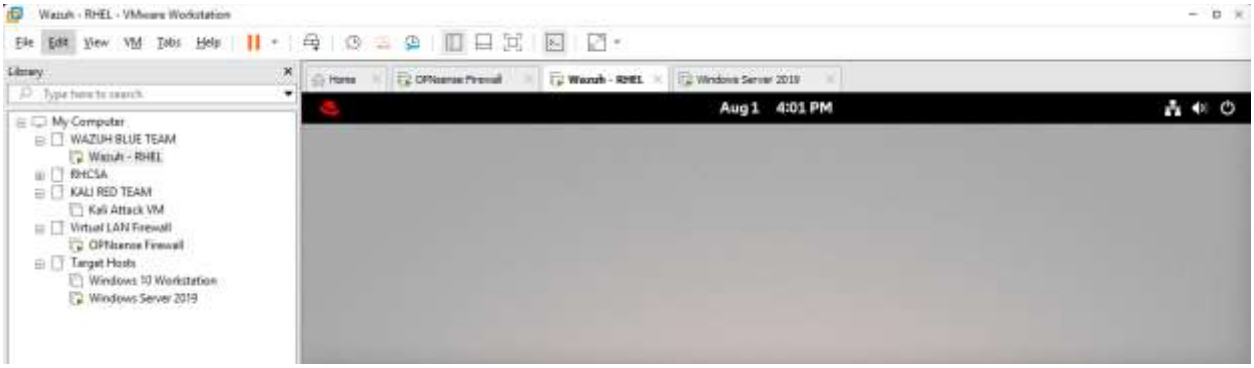
Here is a quick visual representation of the command mentioned above:

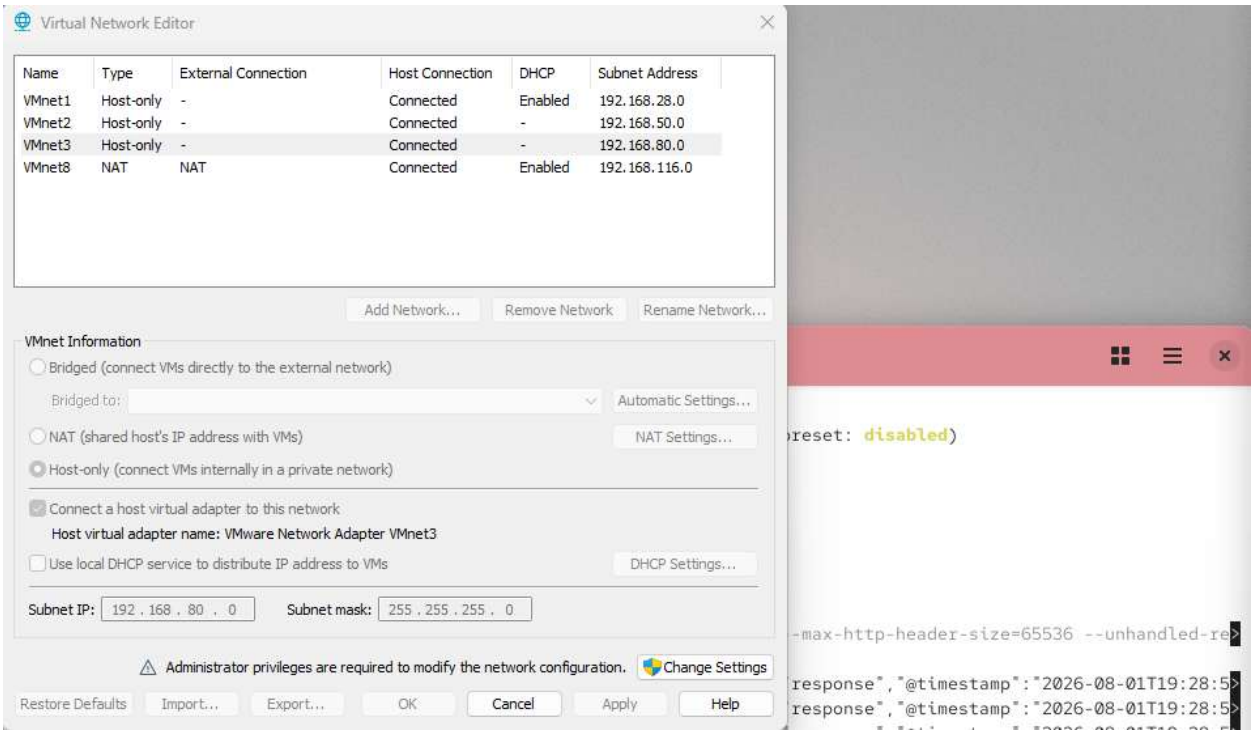


Now that all the initial checks have been completed it is time to use the web UI to deploy agents to the following systems: Windows Server 2019 & Windows 10 Workstation.

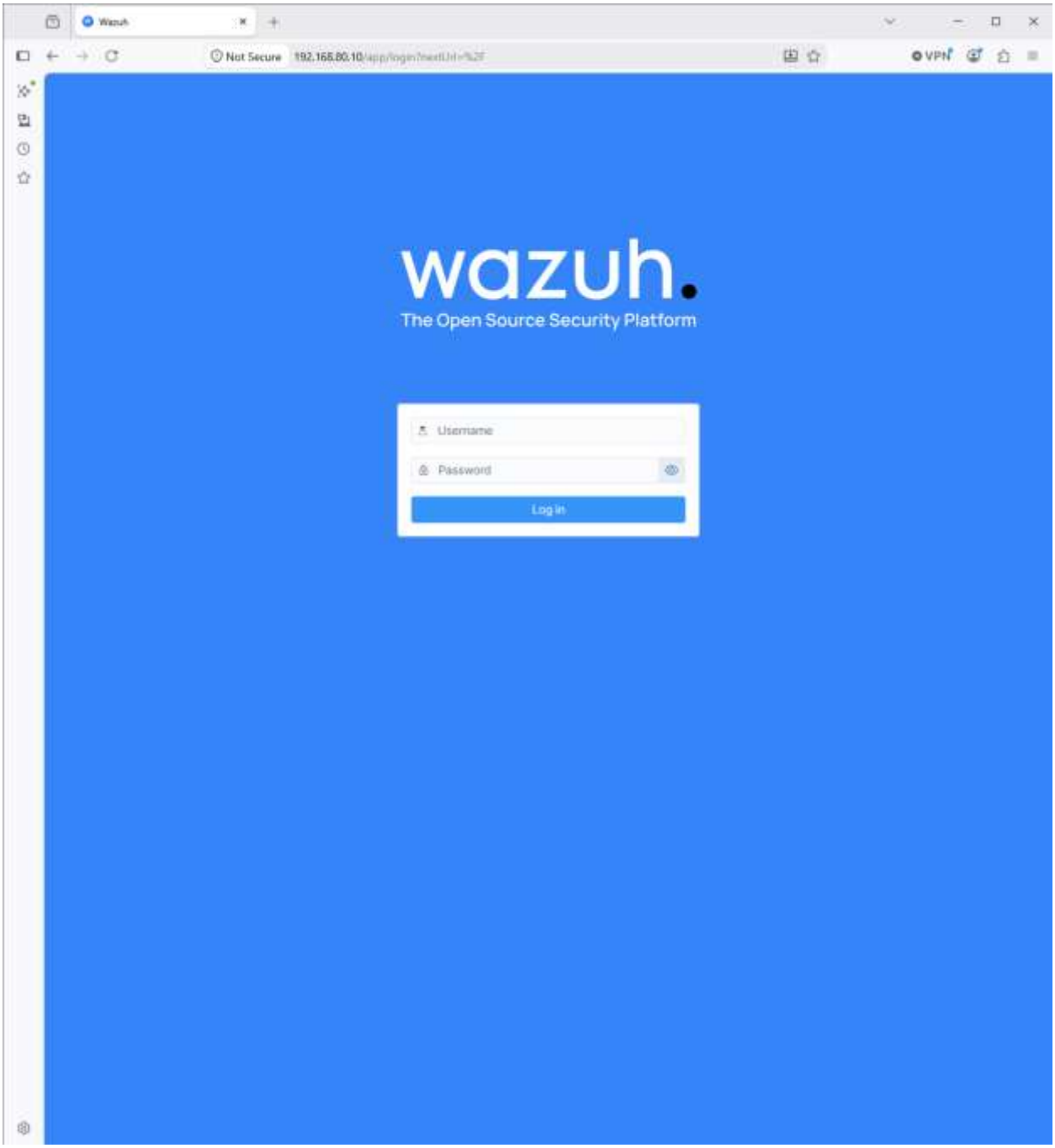
The web UI is located at https://<Wazuh-Machine-IP>:443

Before connecting ensure the setting "***Connect a host virtual adapter to this network***" is enabled on VMware Workstation or you may need to create a port forward rule for the WAN interface on the OPNsense firewall. This enabled configuration will allow direct communication between your host machine and the machine hosting your web UI. You can find the settings in the "***edit"*** tab and selecting "***virtual network editor***". Here is a visual representation:

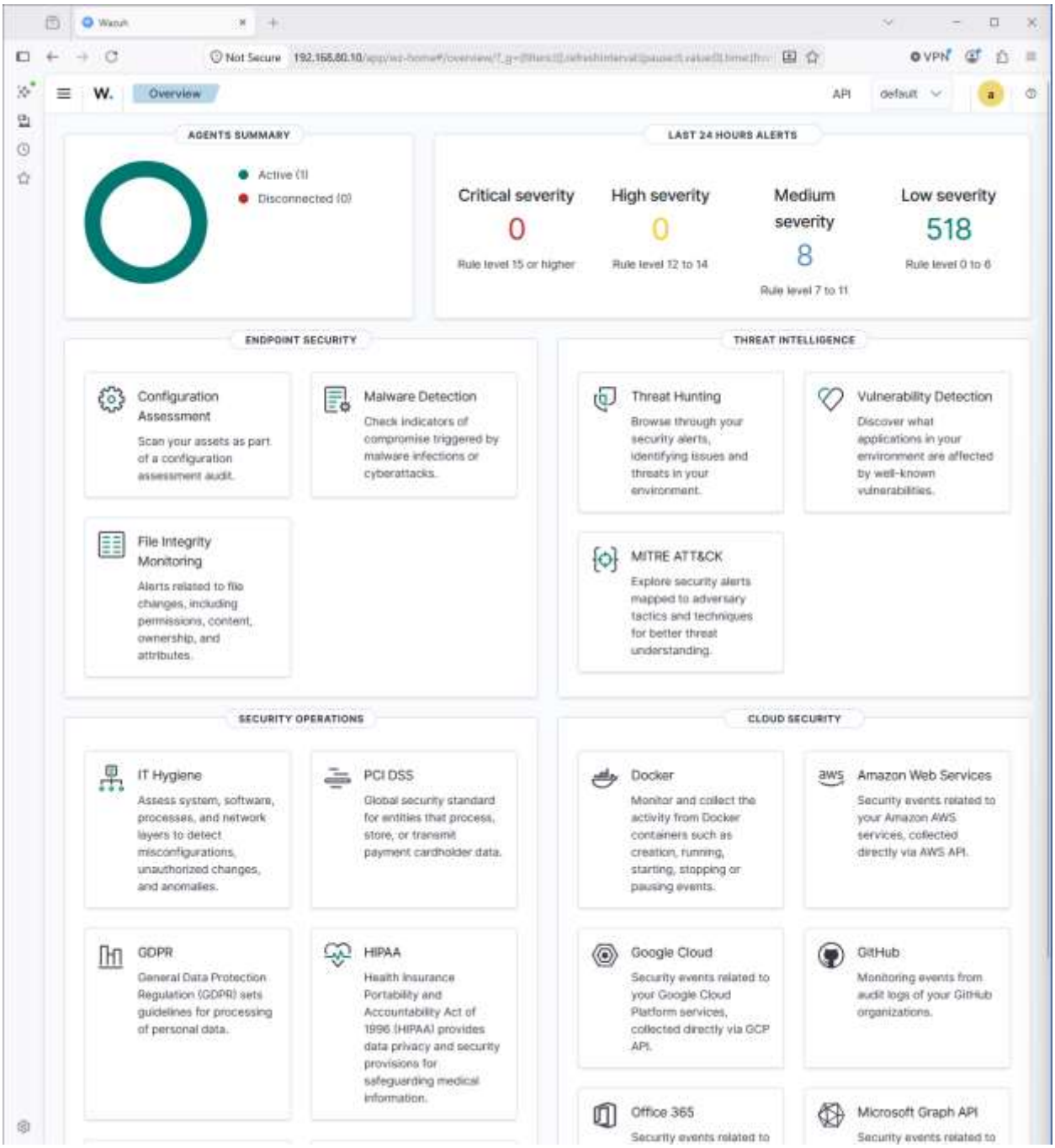




Keep in mind that you will replace the field “Wazuh-Machine-IP” with the IPv4 address of the machine you installed the core services on. Here is the webpage you should see before logging in.

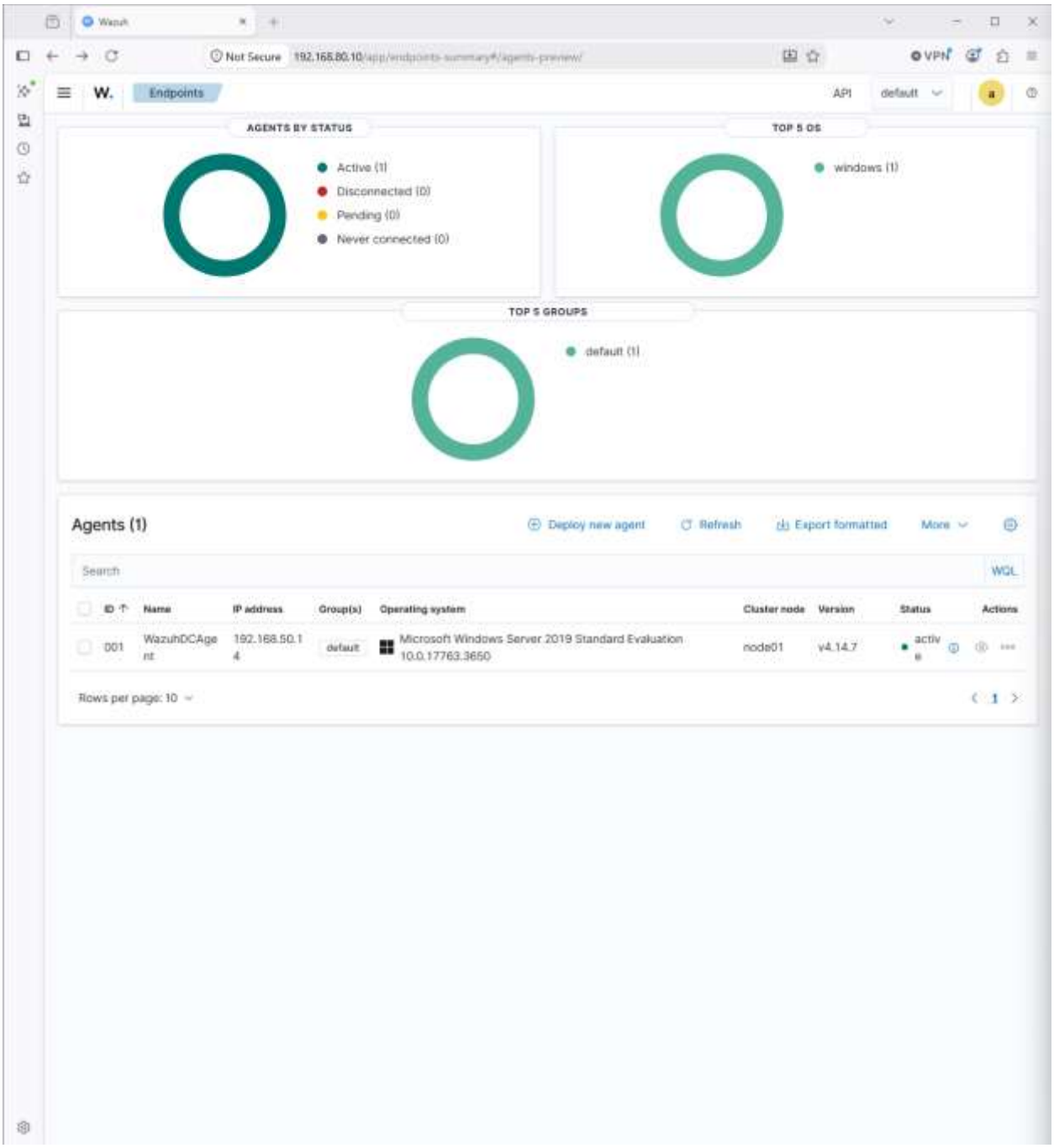


From here I logged in using the administrator credentials generated for me during the phase one setup. Here is the initial dashboard you will see once logged in:

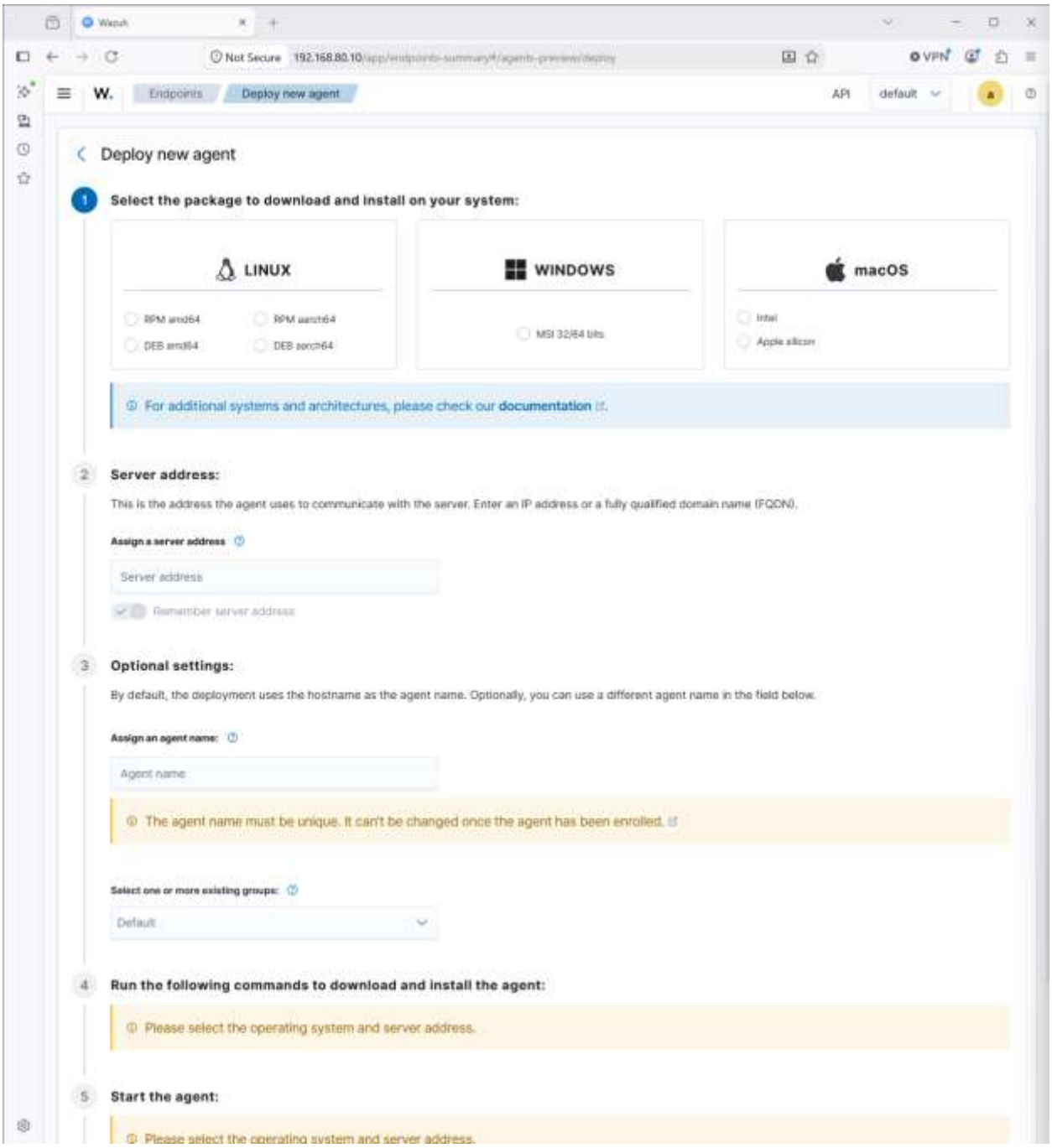


You may notice that I have one agent active in my agent summary. After your initial setup (installing Wazuh and core services), that will not be there, but you will have an option to deploy a new agent which is very straightforward.

I am going to click on the ***active agents*** section on my dashboard and ***deploy new agent*** links.



You will then be presented with this page where you will fill out some information about the system you are deploying the agent to.

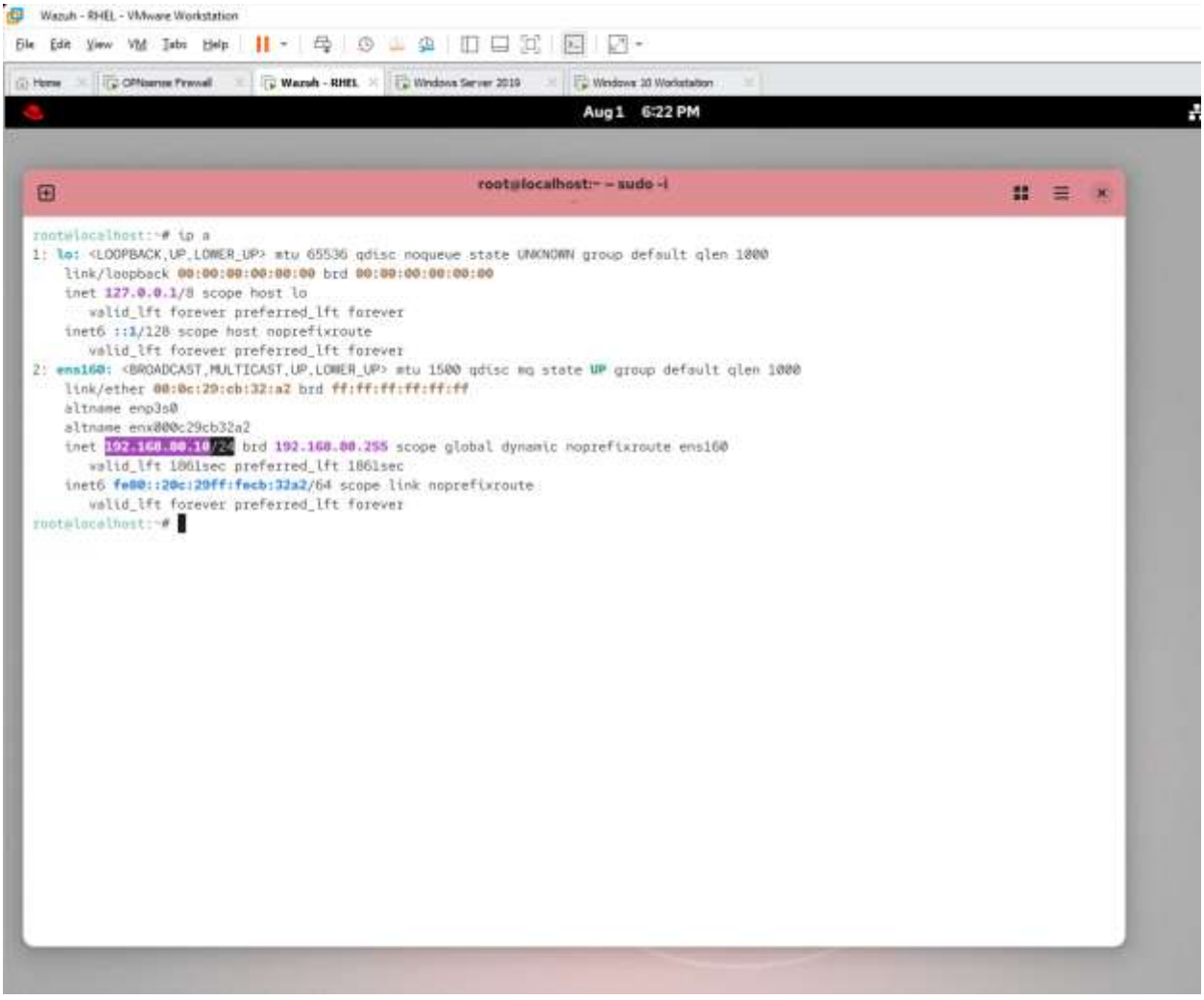


I will be showing my initial deployment for Windows 10 Workstation as an example.

First I will begin by selecting the operating system I am deploying the agent to. This matters because it determines the packages that will be installed on your system to ensure the agent is functioning properly.

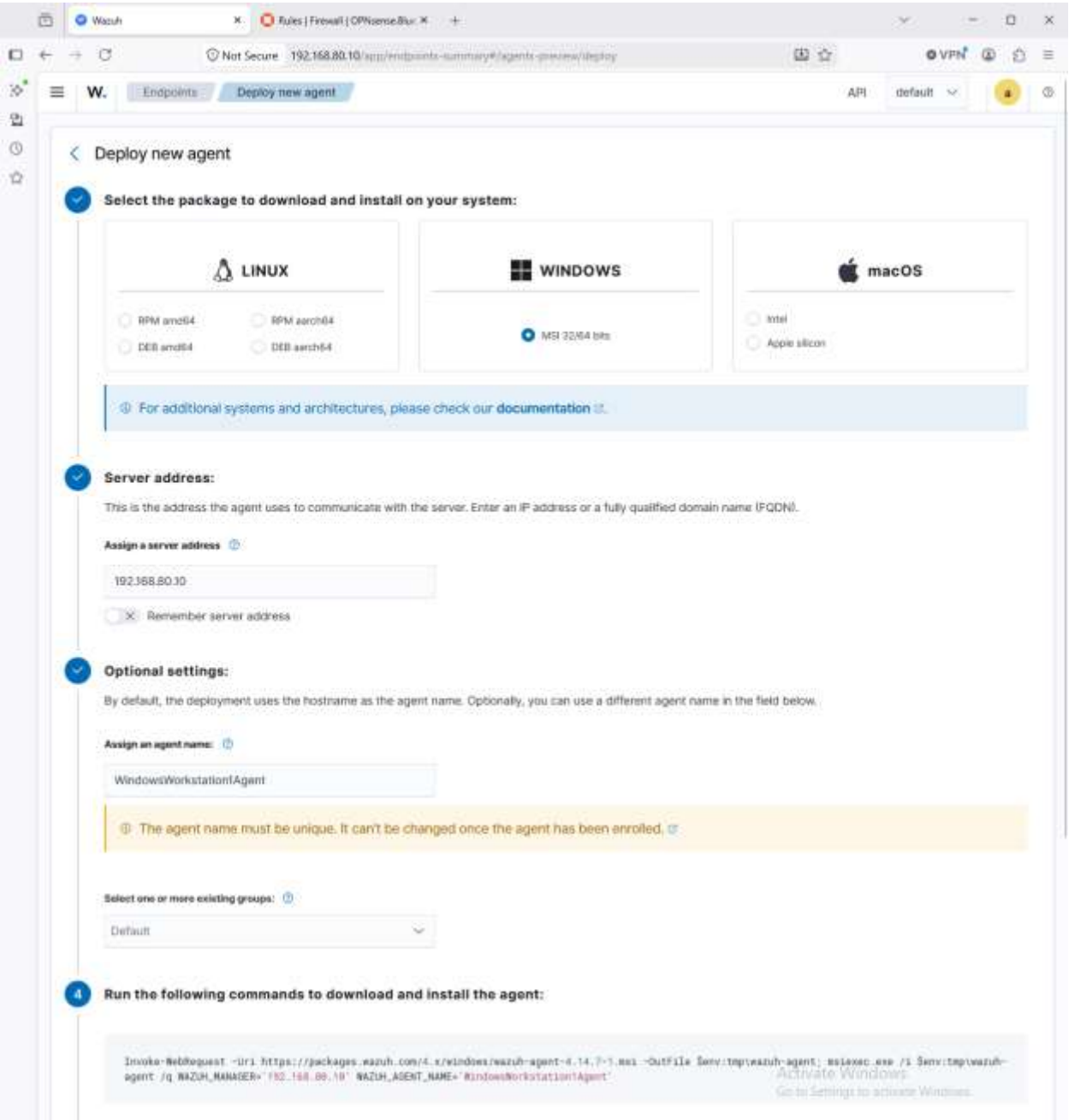
I will then enter the server address or in other words the IPv4 address of the central server which is the RHEL 10 virtual machine.

I will first validate the IP address of the RHEL machine to ensure I am entering the correct information. I will enter the command "***ip a***" into the terminal.



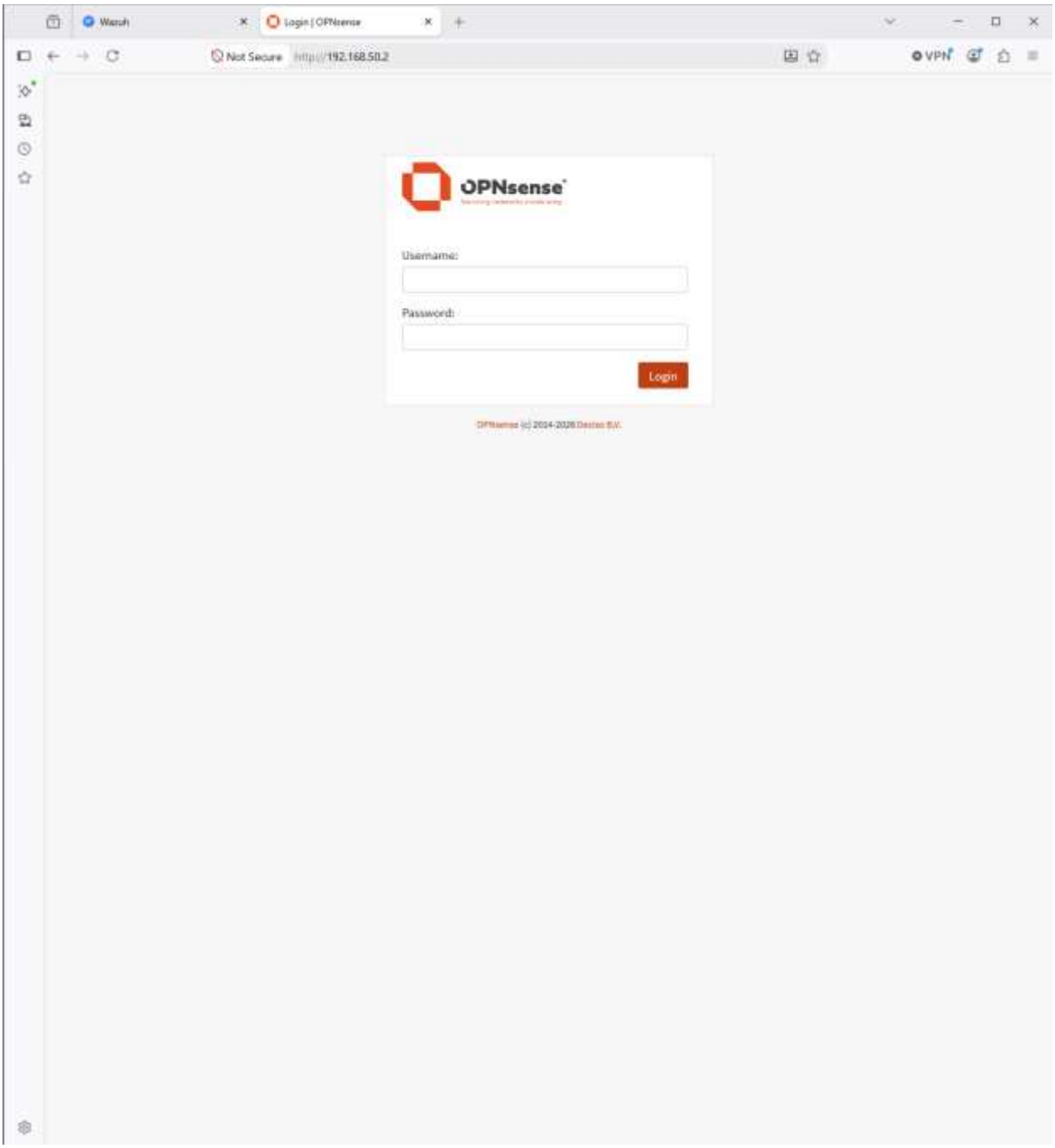
As you can see the IPv4 address I will be using is 192.168.80.10, keep in mind the CIDR notation at the end is not required.

Switching back to the Wazuh web UI we will now enter the address 192.168.80.10 as well as assign an agent name in step three. I will be entering the name WindowsWorkstation1Agent:

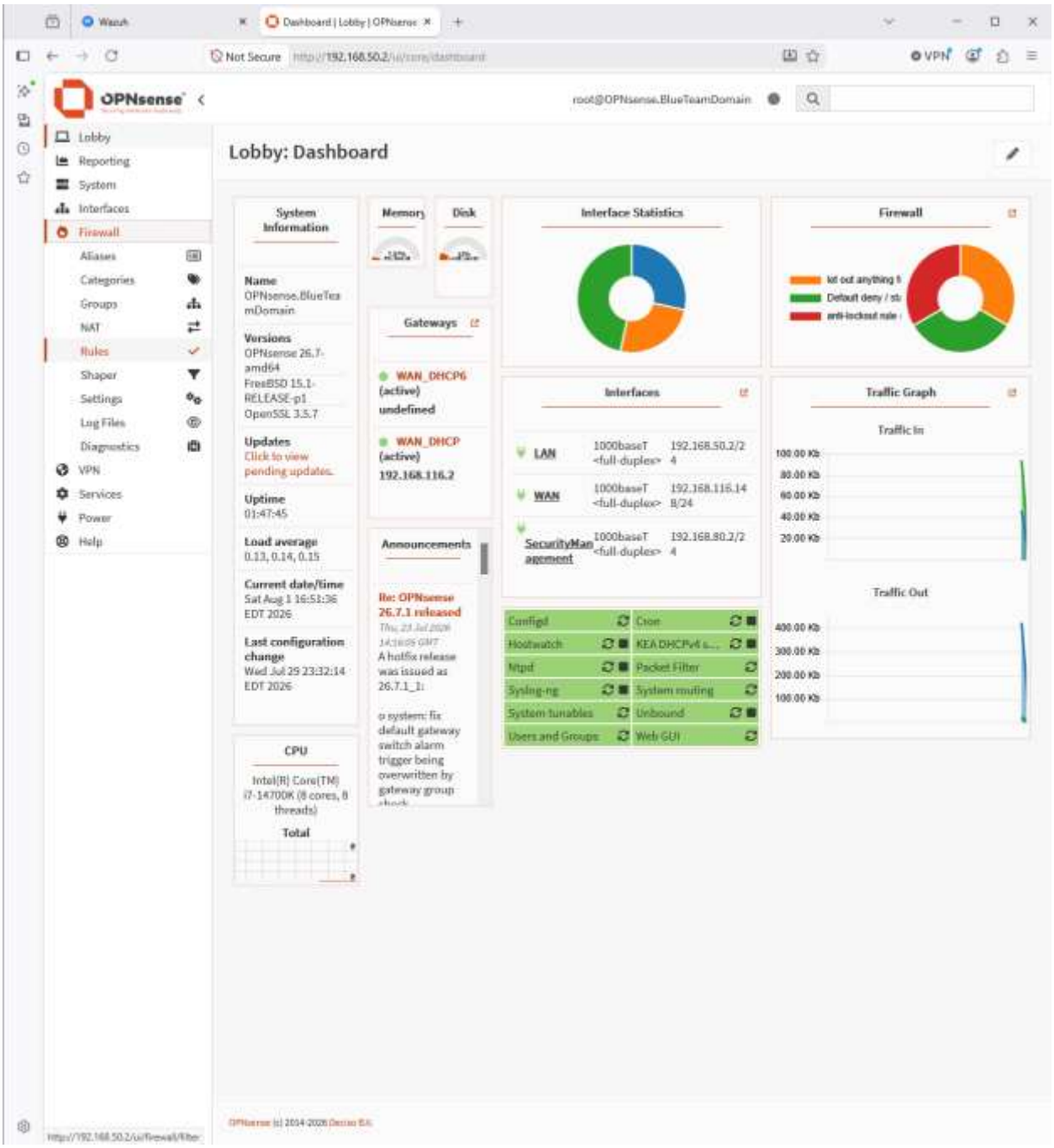


You may leave the optional step as default and below you will see a command that you will be entering in the PowerShell terminal, but I will save that command for later as we need to adjust the firewall rules in OPNsense to allow the agents to send telemetry to the manager.

Moving to the OPNsense web UI:



I will enter my login credentials and head towards the ***firewalls section*** and then ***rules subsection.***



I will then filter my firewall rules as I only need to see rules pertaining to the LAN interface currently.

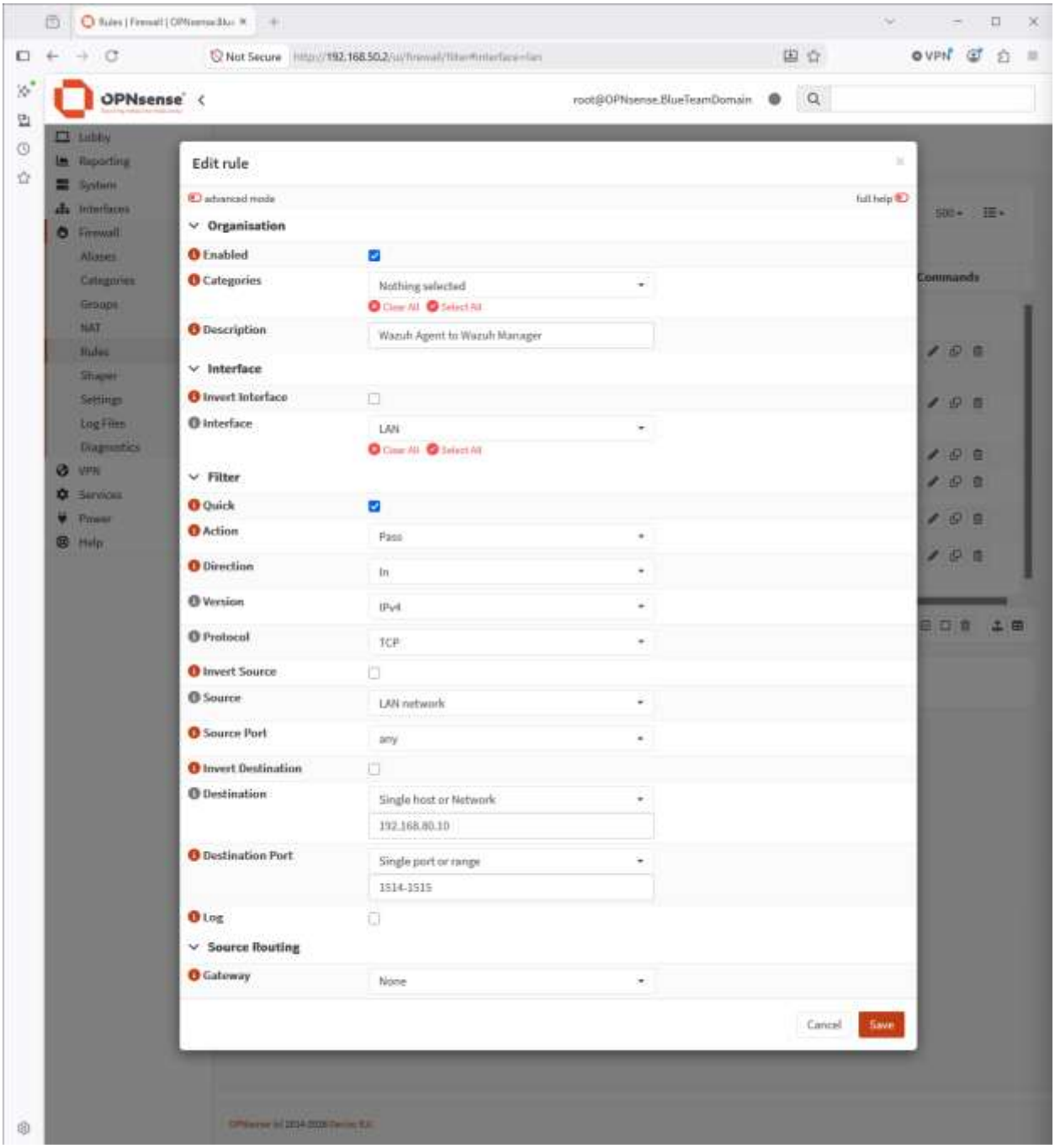
I will then create a rule that does the following:

Allow traffic into the LAN interface from any source IP from the LAN network and ports range 1514 – 1515 to the destination IP 192.168.80.10 on ports range 1514 – 1515. Telemetry being sent requires one hundred percent accuracy; thus, the protocol being used is TCP. Transmission Control Protocol ensures the telemetry will be sent to the destination since any missing packets will be retransmitted.

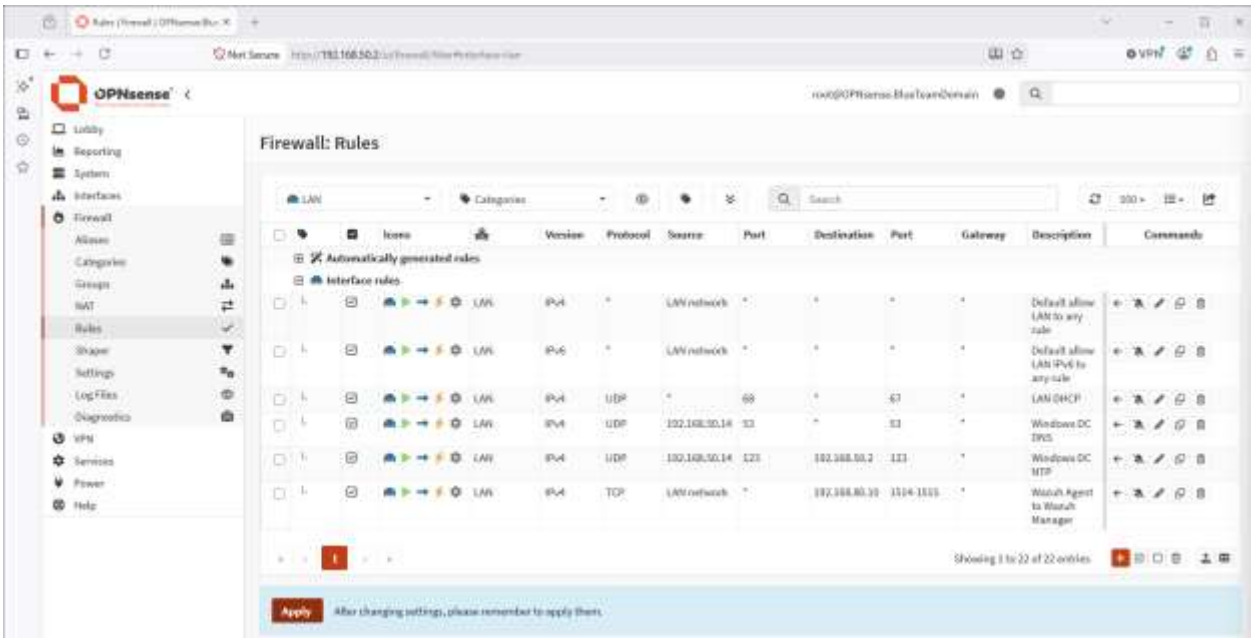
Here is an example:

Interface	Version	Direction	Protocol	Source	Port	Dest	Port	Action
LAN	IPv4	IN	TCP	LAN Network	*	192.168.80.10	1514 - 1515	Allow

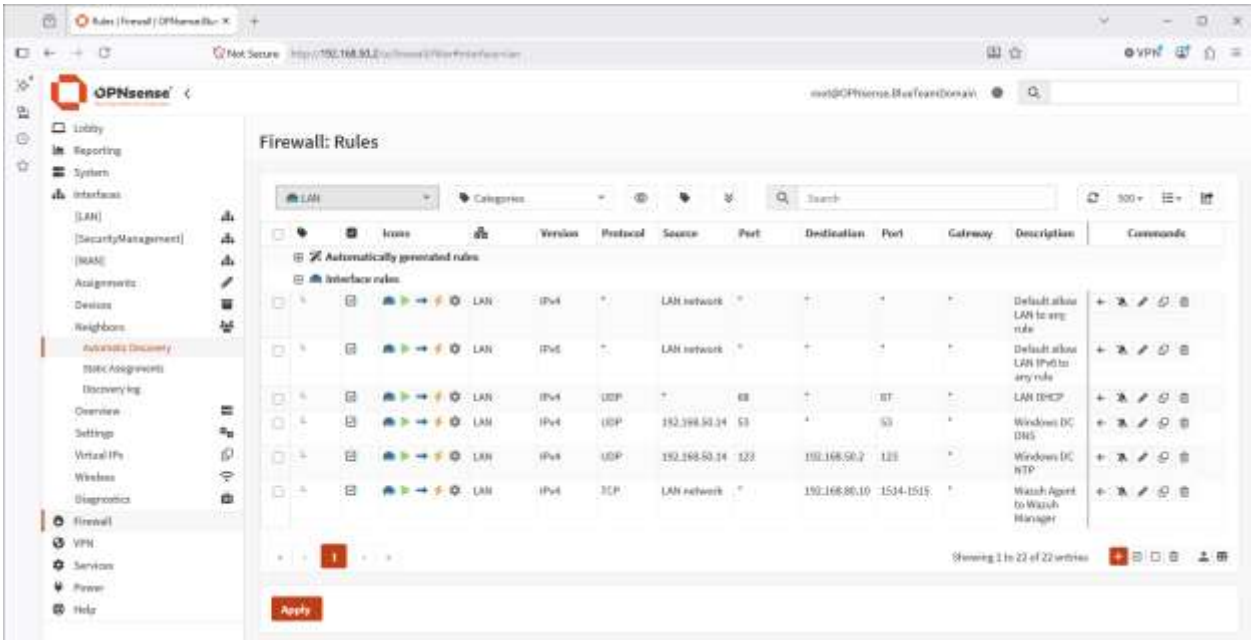
Here is a detailed image of the rule created:



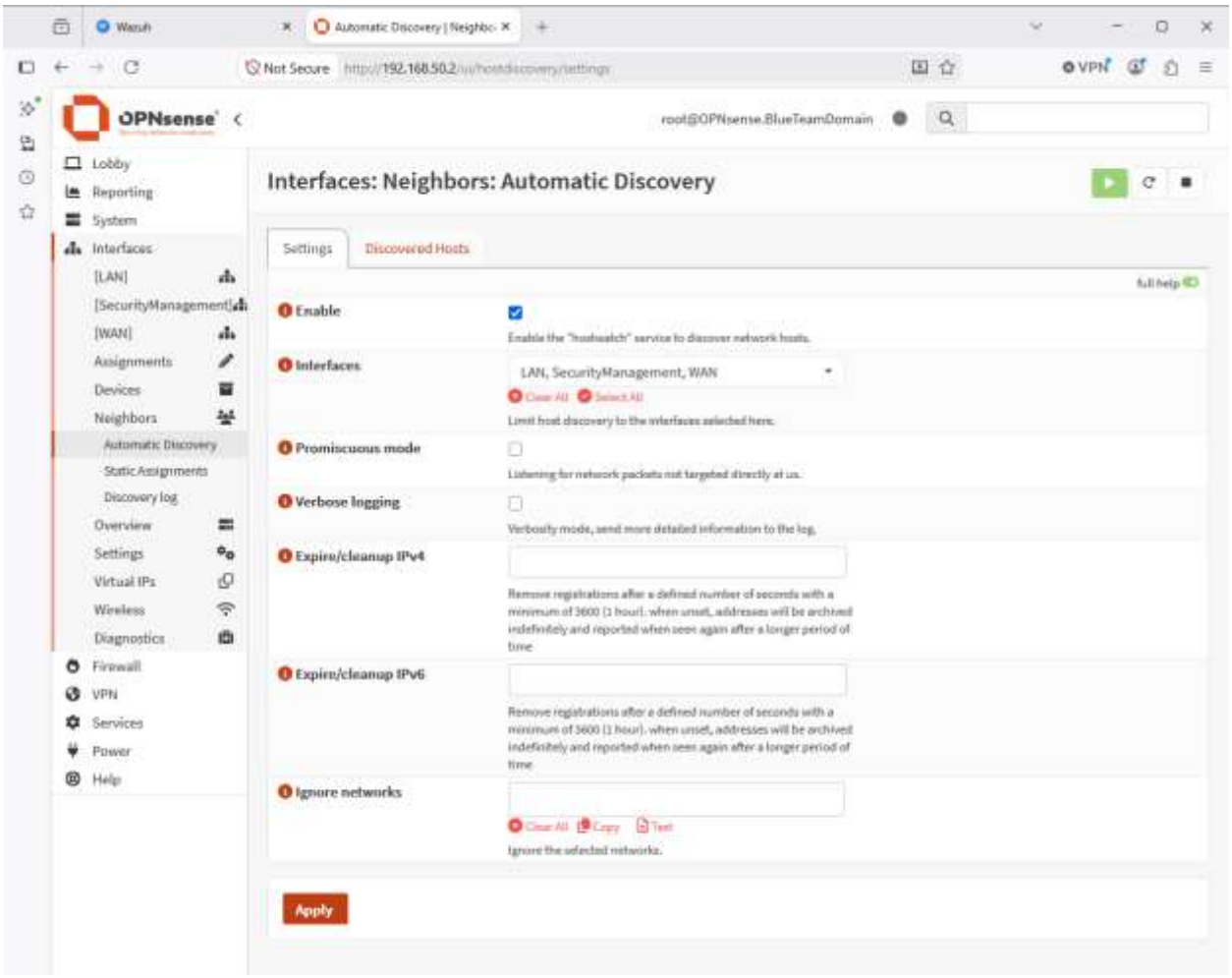
I will now apply the changes to the rule and then verify one more setting within the OPNsense web UI.



Moving on I will now head to the tab labeled ***Interfaces***, select the subsection ***Neighbors***, and the subsection ***automatic discovery***.

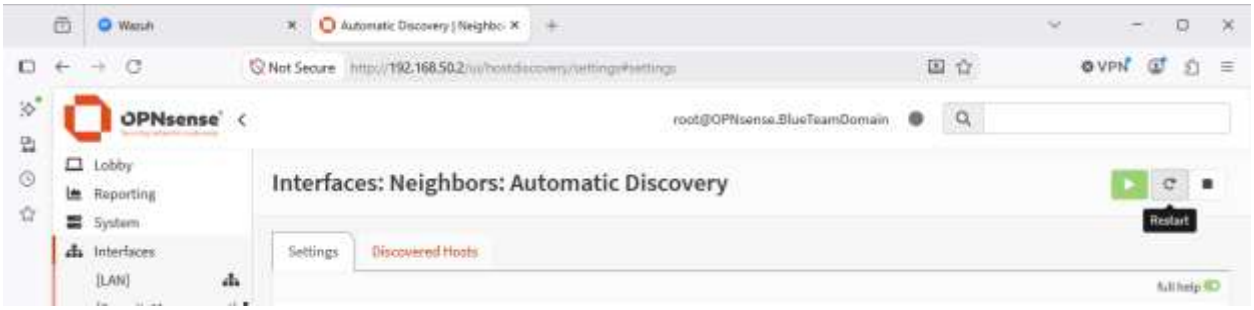


From here I will enable the setting and then in the section labeled interfaces I will select all interfaces (LAN, SecurityManagement, WAN).

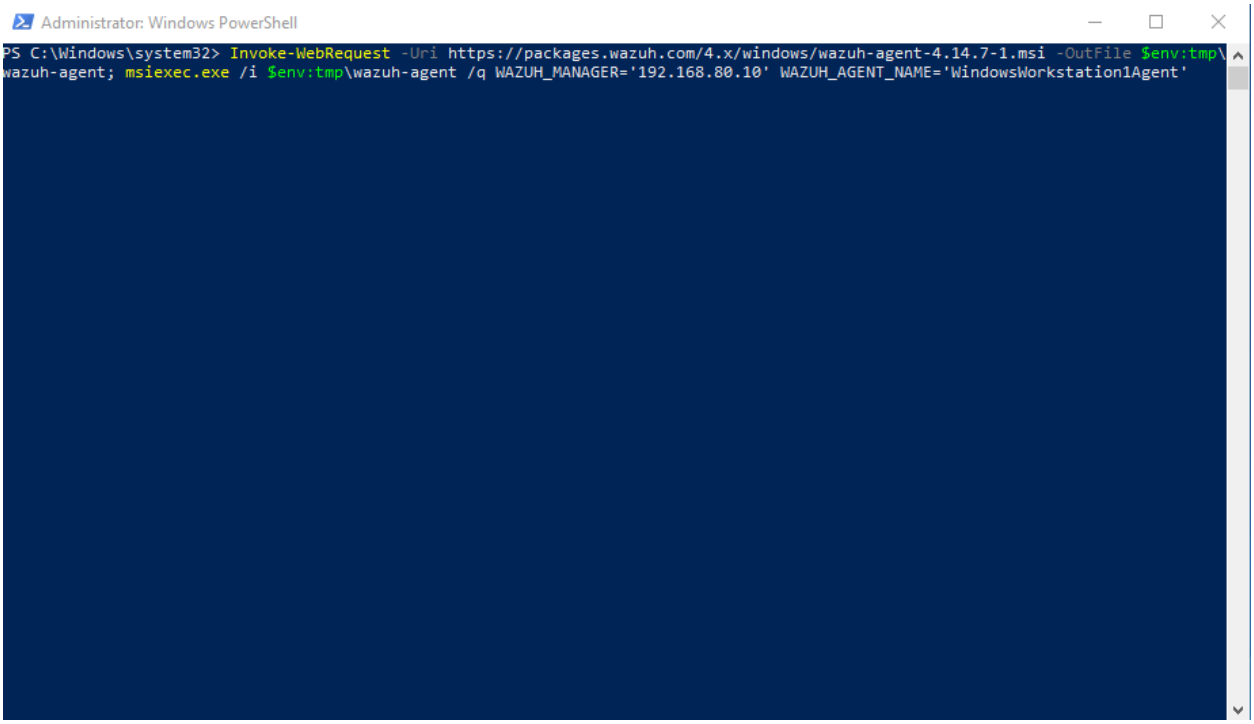


In OPNsense, the **Automatic Discovery** feature was enabled across the active interfaces (LAN, SecurityManagement, and WAN). This service actively tracks ARP and neighbor discovery data across subnets, providing administrative visibility into active endpoints regardless of IP or interface changes. Enabling cross-interface host discovery ensures that endpoints across isolated network segments remain visible within the gateway's management engine.

Once done we will restart the service as shown:



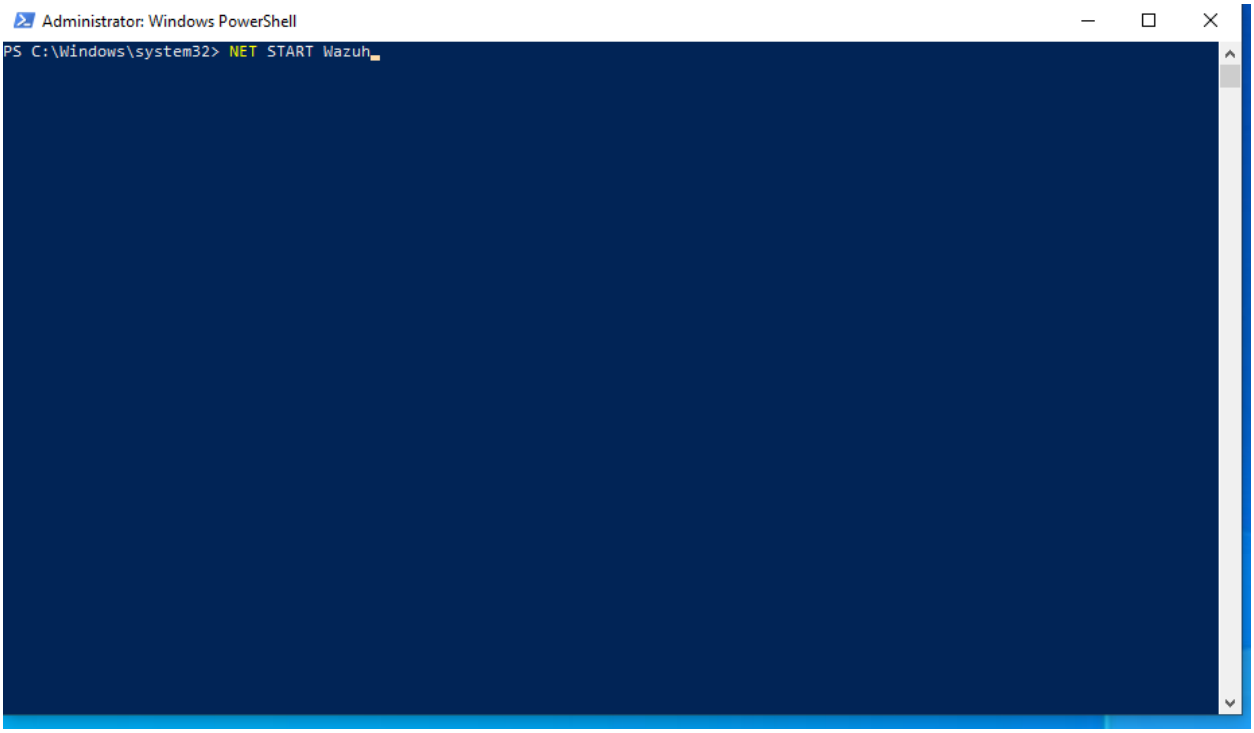
Moving on we will now copy the command generated by the Wazuh web UI and paste it into the PowerShell terminal on the Windows 10 Workstation.



After executing the command, it is now time to start the "***Wazuh***" service.

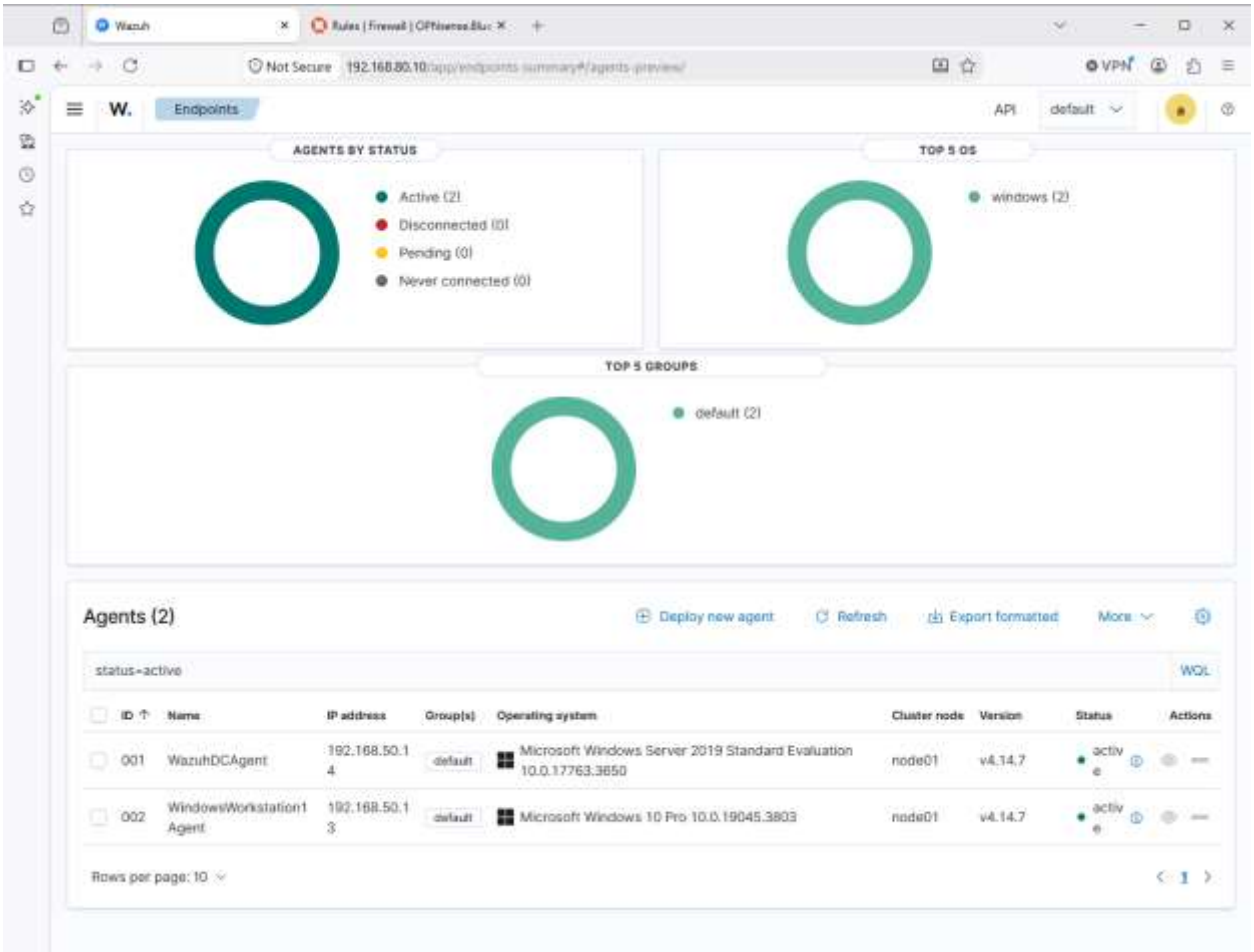
I will be using the command "***NET START Wazuh***".

Starting services requires administrative privileges so ensure you run PowerShell as administrator.



We can now verify the service is running by entering "Get-Service -Name Wazuh", you can see the service is running.

Now we will switch over to the web UI for Wazuh and verify the agent is online and sending telemetry.



As you can see we successfully deployed an agent. Keep in mind that for all this to work correctly these requirements must be met:

- Firewall rule allowing traffic from between two interfaces
- Host-based Firewall on the central server must allow traffic on TCP/1514 & TCP/1515
- All information entered in the web UI to generate the command must be correct
- Ensure the service is started and running
- (Optional) Enable automatic discovery between interfaces

Validate all the requirements listed above are met and the agent should deploy without issues.

This concludes phase two of the SOC home lab project.